

Hayden J Roszell

Denver, CO

hroszell@gmail.com

haydenroszell.com
github.com/m8rmclaren

303.507.7942

ACADEMIA

Bachelor of Science in Software Engineering at Embry-Riddle Aeronautical University

May 2023

Space Track – GPA: 3.89 (Dean's List, Magna cum Laude)

INDUSTRY

Product Owner **Keyfactor, Inc.** – Remote November 2024 – Current

- Product*
- Worked closely with customers to ensure delivery of critical features in Keyfactor's flagship product and several ancillary products.
 - Conducted internal interviews and implemented associated optimizations to resolve major pain points associated with deploying and scaling core product integrations.

Software Engineer - Integrations **Keyfactor, Inc.** – Remote June 2020 – November 2024

- Developer Platform*
- Designed and implemented internal developer platform used by 20%+ of Keyfactor staff to provision demo, dev, and QA environments. Reduced environment setup time from hours to <10 minutes with a Go-based microservice implementing a declarative REST API.
 - Supported 100+ concurrent environments across 40+ Kubernetes nodes, handling 1.8M+ daily HTTP requests and facilitating ingestion of 68M+ log records per day (ELK stack).
- Product Integration*
- Authored product integrations for **X509 Certificate** Orchestration to services including Azure Entra ID (service principals) and Azure Application Gateway.
 - Authored product integrations to extend X509 Certificate enrollment to Certificate Authorities including GoDaddy and GCP CAS.
- DevOps*
- Designed and integrated documentation generation stack with Jinja2/Selenium (Python) that increases documentation usability and reduces engineering effort.
 - Implemented HashiCorp **Vault Secrets engine** (Go) and **Terraform Providers** (Go) for core Keyfactor products that extend PKI capabilities to the respective platform.

Service Desk Intern **HomeAdvisor | AngiesList Inc.** – Denver, Colorado July 2018 – August 2019

- Troubleshooting*
- Served as first-line IT support for an organization of over 2,800 at HomeAdvisor.
 - Diagnosed & addressed issues pertaining to Exchange, Citrix, Avaya, among others.
- Teamwork & Communication*
- Corporate communication tools such as Slack used to collaborate with multiple teams for effective resolution of issues, primarily pertaining to sector-wide outages.

ACTIVITIES

Bus Systems Integration Lead **EagleSat** – Prescott, Arizona May 2021 – May 2023

- On-Board Computer*
- **Led team of 6** engineering students to design and implement on-board computer (OBC) architecture (C) that implements satellite operations including attitude control, data collection, telemetry storage and transmission, and power management.
 - Implemented **FreeRTOS** kernel on STM32 MCU for OBC real-time operating system.
 - Worked closely with satellite vendors globally to overcome supply chain shortages.
- Ground Station*
- Designed and constructed **UHF ground station** for communication with satellites in LEO.
 - Designed and implemented communication protocol to implement **C2** with EagleSat-2.

Civil Air Patrol **USAF Auxiliary** – Denver, Colorado October 2014 – August 2020

- Flight Training*
- Awarded a full-ride **USAF scholarship** for Private Pilot flight training. Accumulated 40 hours in a Cirrus SR20 and 23 hours in a Cessna 172 to successfully obtain **Private Pilot's License**.

ADDITIONAL QUALIFICATIONS

- Computer Skills*
- Proficient with Linux systems, Kubernetes/Docker, PKI concepts, modern identity (OAuth2/OIDC), infrastructure as code (Terraform).
- Coding*
- Proficient in C, C++, Java, Go, Python, C# (.NET), PowerShell, Bash.
- Dev Stack*
- Yabai (MacOS)/I3 (Arch) + zsh + tmux + Neovim
- Publications and Research*
- Roszell, H., Sulyman, A., West, H., & Parmenter, J. (2022, September 12). *Communication Architecture for CubeSats*. 2022 WiSEE.
 - Speaker at 2022 and 2023 NASA Space Grant symposium. *EagleSat II: Developing and Integrating Satellite Communications Systems*

